Mathematical Framework for Conditional Mutual Information Estimation

A Cross-Fitted Approach for Comparing Predictive Targets

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1 Introduction

This document presents the complete mathematical framework for estimating conditional mutual information (CMI) to compare predictive targets in binary classification. The methodology provides a principled, information-theoretic approach that quantifies the unique predictive information one score provides about a target beyond what is already captured by conditioning variables.

The goal of this framework is to measure the **unique predictive information** a score S provides about a binary label Y, given some side information Z. In our application, S represents the predictions (probabilities) from a classification model, and Z can be other model predictions, features, or even alternative target definitions.

2 Theoretical Foundation

2.1 Conditional Mutual Information

Definition 1 (Conditional Mutual Information). *For random variables S, Y, and Z, the conditional mutual information is defined as:*

$$I(S; Y \mid Z) = \mathbb{E}\left[\log \frac{p(Y \mid S, Z)}{p(Y \mid Z)}\right] \tag{1}$$

where the expectation is taken over the joint distribution of (S, Y, Z).

This quantity measures the reduction in uncertainty about Y when S is observed, given that Z is already known.

2.2 Entropy Decomposition

Proposition 1 (Entropy Reduction Identity). *The conditional mutual information can be expressed as:*

$$I(S; Y \mid Z) = H(Y \mid Z) - H(Y \mid S, Z) \tag{2}$$

where $H(\cdot \mid \cdot)$ denotes conditional entropy.

Proof. We start from the definition:

$$\begin{aligned} I(S; Y \mid Z) &= \mathbb{E}_{S,Y,Z}\left[\log \frac{p(Y \mid S, Z)}{p(Y \mid Z)}\right] \\ &= \sum_z P(z) \sum_{s,y} P(s, y \mid z) \left[\log P(Y \mid S, Z) - \log P(Y \mid Z)\right] \\ &= -H(Y \mid S, Z) + H(Y \mid Z) \end{aligned}$$

□

For binary classification where $Y \in \{0, 1\}$, we can express the conditional entropies in terms of the binary entropy function.

Definition 2 (Binary Entropy Function)**.** *The binary entropy function in natural units (nats) is:*

$$H_b(p) = -p \log p - (1-p)\log(1-p) \tag{3}$$

for $p \in (0, 1)$*.*

Theorem 1 (CMI via Binary Entropy Reduction)**.** *For binary* Y*, the conditional mutual information can be written as:*

$$\begin{aligned} I(S; Y \mid Z) &= \mathbb{E}_Z\big[H_b(p(Y=1 \mid Z))\big] - \mathbb{E}_{S,Z}\big[H_b(p(Y=1 \mid S, Z))\big] && (4) \\ &= \mathbb{E}\big[H_b(p(Y=1 \mid Z)) - H_b(p(Y=1 \mid S, Z))\big] && (5) \end{aligned}$$

Proof. By the entropy reduction identity,

$$I(S; Y \mid Z) = H(Y \mid Z) - H(Y \mid S, Z)$$

For binary Y, conditional entropy can be written using the binary entropy function:

$$\begin{aligned} H(Y \mid Z) &= \mathbb{E}_Z\left[H_b(p(Y=1 \mid Z))\right] \\ H(Y \mid S, Z) &= \mathbb{E}_{S,Z}\left[H_b(p(Y=1 \mid S, Z))\right] \end{aligned}$$

Subtracting these expressions gives the result. □

Interpretation: CMI measures the average reduction in uncertainty about a binary target Y when adding S to the conditioning set Z, and this uncertainty is measured using the binary entropy function.

3 Estimation via Cross-Entropy Reduction

3.1 Probabilistic Model Approach

We estimate the conditional probabilities $p(Y = 1 \mid Z)$ and $p(Y = 1 \mid S, Z)$ using probabilistic classifiers:

$$\begin{aligned} \hat{p}_0(z) &\approx p(Y=1 \mid Z=z) && (6) \\ \hat{p}_1(s, z) &\approx p(Y=1 \mid S=s, Z=z) && (7) \end{aligned}$$

3.2 Empirical CMI Estimator

Definition 3 (Empirical CMI Estimator)**.** *Given a dataset* $\{(Y_i, S_i, Z_i)\}_{i=1}^n$*, the empirical CMI estimate is:*

$$\widehat{I}(S; Y \mid Z) = \frac{1}{n}\sum_{i=1}^{n}\big[H_b(\hat{p}_0(Z_i)) - H_b(\hat{p}_1(S_i, Z_i))\big] \tag{8}$$

3.3 Log-Loss Equivalence

Theorem 2 (Log-Loss Gain Equivalence). *The empirical CMI estimator is equivalent to the average log-loss gain:*

$$\widehat{I}(S;Y \mid Z) = \frac{1}{n}\sum_{i=1}^{n}\left[\ell(Y_i, \hat{p}_0(Z_i)) - \ell(Y_i, \hat{p}_1(S_i, Z_i))\right] \tag{9}$$

where $\ell(y,p) = -y\log p - (1-y)\log(1-p)$ *is the binary cross-entropy loss.*

Proof. For binary $y \in \{0,1\}$ and probability p:

$$\begin{aligned} \ell(y,p) &= -y\log p - (1-y)\log(1-p) && (10)\\ \mathbb{E}[\ell(Y,p) \mid Y] &= -p(Y=1)\log p - p(Y=0)\log(1-p) && (11)\\ &= -p(Y=1)\log p - (1-p(Y=1))\log(1-p) && (12)\\ &= H_b(p(Y=1)) && (13) \end{aligned}$$

Thus, the expected log-loss equals the binary entropy, establishing the equivalence. □

4 Cross-Fitting for Unbiased Estimation

4.1 The Overfitting Problem

Direct application of the empirical estimator on the same data used to train the auxiliary models $\hat{p}_0$ and $\hat{p}_1$ introduces optimistic bias due to overfitting.

4.2 Cross-Fitting Solution

Algorithm 1 Cross-Fitted CMI Estimation

Require: Dataset $\mathcal{D} = \{(Y_i, S_i, Z_i)\}_{i=1}^{n}$, number of folds K

Partition indices $\{1, \ldots, n\}$ into K disjoint folds $\mathcal{F}_1, \ldots, \mathcal{F}_K$
for $k = 1$ to K **do**
  $\mathcal{D}^{(k)}_{\text{train}} = \{(Y_i, S_i, Z_i) : i \notin \mathcal{F}_k\}$
  $\mathcal{D}^{(k)}_{\text{test}} = \{(Y_i, S_i, Z_i) : i \in \mathcal{F}_k\}$
  Train $\mathcal{M}^{(k)}_{Z}$ on $\{(Y_i, Z_i) : i \notin \mathcal{F}_k\}$
  Train $\mathcal{M}^{(k)}_{SZ}$ on $\{(Y_i, [S_i, Z_i]) : i \notin \mathcal{F}_k\}$
  for $i \in \mathcal{F}_k$ **do**
    $\hat{p}^{(k)}_0(Z_i) = \mathcal{M}^{(k)}_Z(Z_i)$
    $\hat{p}^{(k)}_1(S_i, Z_i) = \mathcal{M}^{(k)}_{SZ}([S_i, Z_i])$
  end for
  $\Delta H_k = \frac{1}{|\mathcal{F}_k|}\sum_{i\in\mathcal{F}_k}\left[H_b(\hat{p}^{(k)}_0(Z_i)) - H_b(\hat{p}^{(k)}_1(S_i, Z_i))\right]$
end for
$\widehat{I}_{\text{CF}}(S;Y \mid Z) = \frac{1}{K}\sum_{k=1}^{K}\Delta H_k$

4.3 Theoretical Properties

Theorem 3 (Asymptotic Unbiasedness). *Under regularity conditions, the cross-fitted estimator* $\widehat{I}_{CF}(S;Y \mid Z)$ *is asymptotically unbiased:*

$$\lim_{n\to\infty}\mathbb{E}[\widehat{I}_{CF}(S;Y \mid Z)] = I(S;Y \mid Z) \tag{14}$$

5 Out-of-Fold Score Generation

5.1 Preventing Score Leakage

To ensure scores S are truly out-of-sample, we generate them using cross-validation:

Algorithm 2 Out-of-Fold Score Generation

Require: Features X, labels Y, classifier $\mathcal{C}$, number of folds K

Initialize score vector $S \in \mathbb{R}^n$
Partition indices into K folds $\mathcal{G}_1, \ldots, \mathcal{G}_K$
for $k = 1$ to K **do**
  Train $\mathcal{C}^{(k)}$ on $\{(X_i, Y_i) : i \notin \mathcal{G}_k\}$
  for $i \in \mathcal{G}_k$ **do**
    $S_i = \mathcal{C}^{(k)}(X_i)$ {Probability of positive class}
  end for
end for
return S

Lemma 1 (Score Independence). *Each score S_i is generated by a model that never observed (X_i, Y_i) during training, ensuring:*

$$S_i \perp Y_i \mid \{(X_j, Y_j) : j \neq i\} \tag{15}$$

6 Application to Target Comparison

In real-world machine learning systems, especially those involving credit risk, churn prediction, or multi-resolution outcomes, practitioners may face multiple alternative definitions of the target variable. These may correspond to different granularities (e.g., loan-level vs customer-level) or to different objectives (e.g., default vs delinquency).

Let us denote:

- $Y^{(1)}$ — the original or reference target (e.g., loan-level default)
- $Y^{(2)}$ — an alternative or transformed target (e.g., customer-level aggregation)
- $S^{(1)}$ — model predictions trained to estimate $Y^{(1)}$
- $S^{(2)}$ — model predictions trained to estimate $Y^{(2)}$

To compare these models and targets, we define several conditional mutual information quantities, each answering a specific question:

6.1 Self-Information

Self-information measures how well each score predicts its own target:

$$I(S^{(1)}; Y^{(1)}) = \mathbb{E}\big[H_b(p(Y^{(1)} = 1)) - H_b(p(Y^{(1)} = 1 \mid S^{(1)}))\big] \tag{16}$$

$$I(S^{(2)}; Y^{(2)}) = \mathbb{E}\big[H_b(p(Y^{(2)} = 1)) - H_b(p(Y^{(2)} = 1 \mid S^{(2)}))\big] \tag{17}$$

These values represent how much uncertainty the score resolves about its own target. High self-information implies strong predictive performance.

6.2 Cross-Target Conditional Mutual Information

We can assess whether a score trained on one target provides predictive signal about another target:

$$I(S^{(1)}; Y^{(2)} \mid Y^{(1)}) = \mathbb{E}\big[H_b(p(Y^{(2)} = 1 \mid Y^{(1)})) - H_b(p(Y^{(2)} = 1 \mid S^{(1)}, Y^{(1)}))\big] \tag{18}$$
$$I(S^{(2)}; Y^{(1)} \mid Y^{(2)}) = \mathbb{E}\big[H_b(p(Y^{(1)} = 1 \mid Y^{(2)})) - H_b(p(Y^{(1)} = 1 \mid S^{(2)}, Y^{(2)}))\big] \tag{19}$$

These quantities answer: "Does $S^{(1)}$ help predict $Y^{(2)}$ beyond what is known from $Y^{(1)}$?" and vice versa.

6.3 Unique Score Contribution

To measure the unique contribution of one score when another is already known, we compute:

$$I(S^{(1)}; Y^{(2)} \mid S^{(2)}) = \mathbb{E}\big[H_b(p(Y^{(2)} = 1 \mid S^{(2)})) - H_b(p(Y^{(2)} = 1 \mid S^{(1)}, S^{(2)}))\big] \tag{20}$$
$$I(S^{(2)}; Y^{(1)} \mid S^{(1)}) = \mathbb{E}\big[H_b(p(Y^{(1)} = 1 \mid S^{(1)})) - H_b(p(Y^{(1)} = 1 \mid S^{(2)}, S^{(1)}))\big] \tag{21}$$

These terms allow us to evaluate whether one score contains additional predictive information that is not captured by the other.

6.4 Interpretation and Use

This multi-metric comparison provides a principled way to answer key modeling questions:

- **Which model is better at predicting its own target?** *(Use self-information)*
- **Does a model trained on one target generalize to another target?** *(Use cross-target CMI)*
- **Is a score redundant with another, or does it provide complementary signal?** *(Use unique score contribution)*

These measurements are particularly useful when considering switching target definitions, merging modeling pipelines, or evaluating whether multi-task learning may be beneficial.

6.5 Clarifications on Generalization vs Complementarity

While CMI-based metrics allow us to probe the predictive relationships between models and targets, it is important to clarify what each quantity captures:

- **AUC vs CMI:** Traditional metrics like AUC can assess how well a model trained on target $Y^{(1)}$ performs directly on $Y^{(2)}$. However, CMI provides a more granular, information-theoretic view: it measures the reduction in uncertainty of $Y^{(2)}$ when knowing $S^{(1)}$, conditioned on another variable (e.g., $S^{(2)}$ or $Y^{(1)}$). Thus, CMI is not a replacement for AUC but a complementary diagnostic.
- **Complementarity, not just generalization:** A high value of $I(S^{(1)}; Y^{(2)} \mid S^{(2)})$ does not imply that $S^{(1)}$ is a strong standalone predictor of $Y^{(2)}$. Rather, it means that $S^{(1)}$ provides additional signal beyond $S^{(2)}$. This is a measure of *complementarity*, not conventional generalization.

- **Weak models can be informative:** A model with low predictive power on its own task may still have high conditional CMI with respect to another target. This implies that it captures some orthogonal structure useful for a secondary prediction task. Such models may be valuable in stacked ensembles, multi-view learning[1], or representation learning[2].

These distinctions are important in practice: while performance metrics guide deployment, CMI helps us understand model *usefulness* in richer model compositions and diagnostic analyses.

7 Numerical Implementation

7.1 Numerical Stability

Definition 4 (Clipped Binary Entropy)**.** *To avoid numerical issues at boundaries, we implement:*

$$H_b^\epsilon(p) = H_b(\mathit{clip}(p, \epsilon, 1-\epsilon)) \tag{22}$$

where $\mathit{clip}(p, a, b) = \max(a, \min(p, b))$ *and* $\epsilon = 10^{-12}$.

7.2 Variance Estimation

The standard deviation across cross-fitting folds provides uncertainty quantification:

$$\mathrm{SE}(\widehat{I}_{\mathrm{CF}}) = \sqrt{\frac{1}{K-1}\sum_{k=1}^{K}(\Delta H_k - \bar{\Delta H})^2} \tag{23}$$

where $\bar{\Delta H} = \frac{1}{K}\sum_{k=1}^{K}\Delta H_k$.

8 Interpretation and Units

8.1 Information-Theoretic Interpretation

- $I(S; Y \mid Z) > 0$: Score S provides unique information about Y beyond Z
- $I(S; Y \mid Z) \approx 0$: Score S is conditionally redundant given Z
- $I(S; Y \mid Z) < 0$: Theoretically impossible for well-calibrated models

8.2 Units and Scaling

All quantities are measured in **nats** (natural logarithm base). To convert to bits:

$$I_{\mathrm{bits}} = \frac{I_{\mathrm{nats}}}{\log 2} \tag{24}$$

[1]Multi-view learning refers to machine learning methods that combine multiple representations (or ”views”) of the same data, often from different sources or feature sets, to improve prediction.

[2]Representation learning involves training models to automatically discover useful features or representations from raw data, often with the goal of improving performance on downstream tasks.

9 Assumptions and Limitations

9.1 Model Assumptions

1. **Auxiliary Model Accuracy**: CMI estimates depend on the quality of probability estimates from $\mathcal{M}_Z$ and $\mathcal{M}_{SZ}$.

2. **Calibration**: Models should provide well-calibrated probability estimates.

3. **Sufficient Sample Size**: Asymptotic properties require large n relative to model complexity.

9.2 Finite Sample Considerations

Theorem 4 (Finite Sample Bias)**.** *For finite samples and imperfect auxiliary models, the cross-fitted estimator may exhibit bias:*

$$\mathbb{E}[\widehat{I}_{CF}(S; Y \mid Z)] = I(S; Y \mid Z) + bias(n, K, \mathcal{M}) \tag{25}$$

where the bias depends on sample size n, number of folds K, and model class $\mathcal{M}$.

10 Conclusion

This mathematical framework provides a rigorous foundation for comparing predictive targets using conditional mutual information. The cross-fitting approach ensures unbiased estimation while the entropy reduction formulation offers computational tractability. The methodology extends beyond traditional performance metrics like AUC to quantify the unique predictive content captured by different target definitions.